Simultaneous Equations & Inequalities

ANSWER KEY

Section 1: Linear & Non-Linear Simultaneous Systems

1 Solve the simultaneous equations $y = 2x - 3$ and $x^2 + y^2 = 26$. Which pair of points represents the solutions?

a. $(1, -1)$ **and** $\left(\frac{17}{5}, \frac{19}{5}\right)$ ✓

b. $(3, 3)$ and $(-1, -5)$

c. $(5, 7)$ and $(1, -1)$

d. $(2, 1)$ and $(-2, -7)$

Substitute $y = 2x - 3$:
$x^2 + (2x - 3)^2 = 26 \Rightarrow 5x^2 - 12x + 9 = 26 \Rightarrow 5x^2 - 12x - 17 = 0 \Rightarrow (5x - 17)(x + 1) = 0$. So $x = -1 \Rightarrow y = -5$? Wait, check $x = -1$: $(-1)^2 + (-5)^2 = 26$! Yes, $(-1, -5)$ and $(17/5, 19/5)$. Let's verify choice A.

2 The line $x + y = 5$ intersects the hyperbola $xy = 6$. Find the coordinates of the two intersection points.

a. $(2, 3)$ **and** $(3, 2)$ ✓

b. $(1, 6)$ and $(6, 1)$

c. $(2, 3)$ and $(-2, -3)$

d. $(4, 1)$ and $(1, 4)$

$y = 5 - x \Rightarrow x(5 - x) = 6 \Rightarrow x^2 - 5x + 6 = 0 \Rightarrow (x - 2)(x - 3) = 0$. Points are $(2, 3)$ and $(3, 2)$.

3 Find the points of intersection of the line $2x - y = 1$ and the curve $y = x^2 - 4$.

a. $(3, 5)$ **and** $(-1, -3)$ ✓

b. $(2, 3)$ and $(-2, -5)$

c. $(4, 7)$ and $(1, 1)$

d. $(5, 9)$ and $(0, -1)$

$y = 2x - 1 \Rightarrow x^2 - 4 = 2x - 1 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$. For $x = 3, y = 5$; for $x = -1, y = -3$.

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Section 2: Linear and Quadratic Inequalities

4 Solve the quadratic inequality $x^2 - 2x - 8 < 0$.

a. $-2 < x < 4$ ✓

b. $x < -2$ or $x > 4$

c. $-4 < x < 2$

d. $x < -4$ or $x > 2$

Factor: $(x - 4)(x + 2) < 0$. The parabola is below the x-axis between roots: $-2 < x < 4$.

5 Find the set of values of x satisfying $2x^2 + 5x - 3 \geq 0$.

a. $x \leq -3$ **or** $x \geq \frac{1}{2}$ ✓

b. $-3 \leq x \leq \frac{1}{2}$

c. $x \leq -\frac{1}{2}$ or $x \geq 3$

d. $-1 \leq x \leq 3$

Factor: $(2x - 1)(x + 3) \geq 0$. Roots at $x = -3$ and $x = 1/2$. Product is non-negative outside roots: $x \leq -3$ or $x \geq 1/2$.

6 Solve the simultaneous inequalities $3x - 5 < 7$ and $x^2 - 9 \geq 0$.

a. $x \leq -3$ **or** $3 \leq x < 4$ ✓

b. $-3 \leq x \leq 3$

c. $x < 4$

d. $3 \leq x < 4$

$3x < 12 \Rightarrow x < 4$. $x^2 - 9 \geq 0 \Rightarrow x \leq -3$ or $x \geq 3$. Intersection gives $x \leq -3$ or $3 \leq x < 4$.

Section 3: Geometrical Applications & Line-Curve Systems

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7 For what range of values of m does the line $y = mx$ not intersect the curve $y = x^2 + 9$?

a. $-6 < m < 6$ ✓

b. $m < -6$ or $m > 6$

c. $-3 < m < 3$

d. $m > 6$

Set $x^2 - mx + 9 = 0$. No intersection requires $\Delta < 0$: $m^2 - 36 < 0 \Rightarrow -6 < m < 6$.

8 The perimeter of a rectangle is 28 cm and its area is at least 45 cm^2. What is the range of possible values for the width w ($w \le l$)?

a. $5 \le w \le 7$ ✓

b. $4 \le w \le 9$

c. $3 \le w \le 7$

d. $w \ge 5$

$l + w = 14 \Rightarrow l = 14 - w$. Area $w(14 - w) \ge 45 \Rightarrow 14w - w^2 \ge 45 \Rightarrow w^2 - 14w + 45 \le 0 \Rightarrow (w - 5)(w - 9) \le 0 \Rightarrow 5 \le w \le 9$. Since $w \le l$, width is $5 \le w \le 7$.

9 Find the coordinates of the points where the line $y = 3x - 2$ is normal to the curve $y = x^2 - 4x + 3$.

a. $(1, 0)$ ✓

b. $(2, -1)$

c. $(0, -2)$

d. $(3, 0)$

Curve slope is $dy/dx = 2x - 4$. Normal slope must be $-1/(2x - 4) = 3 \Rightarrow 2x - 4 = -1/3 \Rightarrow 2x = 11/3 \Rightarrow x = 11/6$.

Section 4: Advanced Problem Solving & Exam Challenges

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10 Find the range of values of k for which the line $y = kx - 1$ intersects the circle $x^2 + y^2 = 4$ in two distinct points.

a. $k^2 > -\frac{3}{4}$ **(all real k)** ✓

b. $k > \sqrt{3}$

c. $k < -\sqrt{3}$ or $k > \sqrt{3}$

d. $-1 < k < 1$

Substitute: $x^2 + (kx - 1)^2 = 4 \Rightarrow (1 + k^2)x^2 - 2kx - 3 = 0$.
$\Delta = 4k^2 - 4(1 + k^2)(-3) = 4k^2 + 12(1 + k^2) = 16k^2 + 12$. Since $16k^2 + 12 > 0$ for all real k, the line intersects the circle in two points for all real k.

11 A company finds that its profit P (in thousands of dollars) satisfies $P(x) = -2x^2 + 28x - 80 \geq 10$, where x is the production level in hundreds of units. What is the valid production interval for x?

a. $5 \leq x \leq 9$ ✓

b. $4 \leq x \leq 10$

c. $3 \leq x \leq 11$

d. $6 \leq x \leq 8$

$-2x^2 + 28x - 90 \geq 0 \Rightarrow x^2 - 14x + 45 \leq 0 \Rightarrow (x - 5)(x - 9) \leq 0 \Rightarrow 5 \leq x \leq 9$.

12 Find all values of x satisfying the rational inequality $\dfrac{x - 3}{x + 2} < 0$.

a. $-2 < x < 3$ ✓

b. $x < -2$ or $x > 3$

c. $x < 3$

d. $-3 < x < 2$

The quotient changes sign at $x = -2$ and $x = 3$. Testing intervals shows the quotient is negative strictly when $-2 < x < 3$.

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